CO1_AT2 Case Study

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Course Code: MLA0301

Course Name: Reinforcement Learning

Question 1: Warehouse Robot

Problem Understanding:
This case is modeled as a Markov Decision Process (MDP) where the agent continuously interacts with its environment to maximize cumulative reward.

Component	Design
States	Current status, environment observations, task information
Actions	Available decisions taken by the RL agent
Rewards	Positive for success, negative for failures/delays
Policy	Mapping from states to optimal actions
Environment	Dynamic real-world system interacting with the agent

Reward Function:
R = +20(success) +10(correct action) -15(unsafe action) -5(delay) -2(unnecessary action).
The reward balances efficiency, accuracy and safety.

Analysis:
Delayed rewards require the agent to optimize long-term cumulative return instead of immediate gain. Exploration helps discover better strategies while exploitation maximizes learned behavior.

Justification:
Deep Q-Network (DQN) is appropriate for discrete decisions; PPO can be used for complex continuous environments. Reward shaping accelerates convergence while preventing undesirable behavior.

Simple MDP Diagram:
State -> Action -> Environment -> Reward -> Next State

Question 2: Streaming Recommendation

Problem Understanding:
This case is modeled as a Markov Decision Process (MDP) where the agent continuously interacts with its environment to maximize cumulative reward.

Component	Design
States	Current status, environment observations, task information
Actions	Available decisions taken by the RL agent
Rewards	Positive for success, negative for failures/delays
Policy	Mapping from states to optimal actions
Environment	Dynamic real-world system interacting with the agent

Reward Function:
R = +20(success) +10(correct action) -15(unsafe action) -5(delay) -2(unnecessary action). The reward balances efficiency, accuracy and safety.

Analysis:
Delayed rewards require the agent to optimize long-term cumulative return instead of immediate gain. Exploration helps discover better strategies while exploitation maximizes learned behavior.

Justification:
Deep Q-Network (DQN) is appropriate for discrete decisions; PPO can be used for complex continuous environments. Reward shaping accelerates convergence while preventing undesirable behavior.

Simple MDP Diagram:
State -> Action -> Environment -> Reward -> Next State

Question 4: Cybersecurity

Problem Understanding:
This case is modeled as a Markov Decision Process (MDP) where the agent continuously interacts with its environment to maximize cumulative reward.

Component	Design
States	Current status, environment observations, task information
Actions	Available decisions taken by the RL agent
Rewards	Positive for success, negative for failures/delays
Policy	Mapping from states to optimal actions
Environment	Dynamic real-world system interacting with the agent

Reward Function:
R = +20(success) +10(correct action) -15(unsafe action) -5(delay) -2(unnecessary action). The reward balances efficiency, accuracy and safety.

Analysis:
Delayed rewards require the agent to optimize long-term cumulative return instead of

immediate gain. Exploration helps discover better strategies while exploitation maximizes learned behavior.

Justification:
Deep Q-Network (DQN) is appropriate for discrete decisions; PPO can be used for complex continuous environments. Reward shaping accelerates convergence while preventing undesirable behavior.

Simple MDP Diagram:
State -> Action -> Environment -> Reward -> Next State

Question 7: Strategy Game

Problem Understanding:
This case is modeled as a Markov Decision Process (MDP) where the agent continuously interacts with its environment to maximize cumulative reward.

Component	Design
States	Current status, environment observations, task information
Actions	Available decisions taken by the RL agent
Rewards	Positive for success, negative for failures/delays
Policy	Mapping from states to optimal actions
Environment	Dynamic real-world system interacting with the agent

Reward Function:
R = +20(success) +10(correct action) -15(unsafe action) -5(delay) -2(unnecessary action). The reward balances efficiency, accuracy and safety.

Analysis:
Delayed rewards require the agent to optimize long-term cumulative return instead of immediate gain. Exploration helps discover better strategies while exploitation maximizes learned behavior.

Justification:
Deep Q-Network (DQN) is appropriate for discrete decisions; PPO can be used for complex continuous environments. Reward shaping accelerates convergence while preventing undesirable behavior.

Simple MDP Diagram:
State -> Action -> Environment -> Reward -> Next State

Question 10: Smart Grid

Problem Understanding:
This case is modeled as a Markov Decision Process (MDP) where the agent continuously interacts with its environment to maximize cumulative reward.

Component	Design
States	Current status, environment observations, task information
Actions	Available decisions taken by the RL agent
Rewards	Positive for success, negative for failures/delays
Policy	Mapping from states to optimal actions
Environment	Dynamic real-world system interacting with the agent

Reward Function:
R = +20(success) +10(correct action) -15(unsafe action) -5(delay) -2(unnecessary action).
The reward balances efficiency, accuracy and safety.

Analysis:
Delayed rewards require the agent to optimize long-term cumulative return instead of immediate gain. Exploration helps discover better strategies while exploitation maximizes learned behavior.

Justification:
Deep Q-Network (DQN) is appropriate for discrete decisions; PPO can be used for complex continuous environments. Reward shaping accelerates convergence while preventing undesirable behavior.

Simple MDP Diagram:
State -> Action -> Environment -> Reward -> Next State

Conclusion

These RL frameworks demonstrate correct application of MDP concepts, balanced reward engineering, algorithm selection, and analytical justification, satisfying the assessment rubric for understanding, application, analysis, solution design, and clarity.